2.4 Analysis in $\mathbb{R}^n$: Problems

Attempt each problem for at least 30 minutes before reading solutions.md. $C[a,b]$ carries the sup norm unless stated otherwise.

Warm-up

1. Let $f_n(x) = x^n$ on $[0,1]$. Find the pointwise limit, show the convergence is not uniform on $[0,1]$, and show it is uniform on $[0,a]$ for every $a < 1$.

2. Let $f_n(x) = nxe^{-nx^2}$ on $[0,1]$. Show that $f_n \to 0$ pointwise, but $\int_0^1 f_n \to \tfrac{1}{2}$. Is the convergence uniform?

3. Show that $f(x) = \sum_{n\ge 1} \frac{\sin(nx)}{n^2}$ converges uniformly on $\mathbb{R}$ to a continuous function. Does the term-by-term differentiated series converge at $x = 0$? What does this say about applying the term-by-term differentiation theorem?

4. (a) Compute the Jacobian matrix of $f(x,y) = (x^2 y,\ \sin(xy),\ e^{x+y})$.

(b) Show that $\det : M_n(\mathbb{R}) \to \mathbb{R}$ is differentiable at I with $D(\det)(I)H = \operatorname{tr} H$.

5. Let $f(x,y) = \dfrac{xy}{x^2+y^2}$ for $(x,y) \neq (0,0)$ and $f(0,0) = 0$. Show that both partial derivatives of f exist at $(0,0)$, but f is not differentiable there.

6. Let $U \subseteq \mathbb{R}^n$ be open and connected and $f : U \to \mathbb{R}^m$ differentiable with $Df(x) = 0$ for all $x \in U$. Show that f is constant.

Core

7. Let $\mathcal{F} = \{f \in C^1[0,1] : f(0) = 0,\ |f'(x)| \le 1 \text{ for all } x\}$. Show that $\mathcal{F}$ has compact closure in $C[0,1]$, but that $\mathcal{F}$ itself is not closed.

8. Show that the sequence $f_n(x) = \sin(nx)$ in $C[0,\pi]$ is bounded but has no uniformly convergent subsequence. Which hypothesis of Arzelà–Ascoli fails?

9. Let $f \in C[0,1]$ satisfy $\int_0^1 f(x)x^n\,dx = 0$ for all $n \ge 0$. Show that $f = 0$.

10. Show that polynomials in x^2 (that is, $\{p(x^2) : p \text{ a polynomial}\}$) are dense in $C[0,1]$ but not in $C[-1,1]$.

11. Show that the system

$$x = \tfrac{1}{4}\sin(x+y), \qquad y = \tfrac{1}{4}\cos(x-y)$$

has exactly one solution $(x,y) \in \mathbb{R}^2$.

12. Let $F : [0,1] \times \mathbb{R} \to \mathbb{R}$ be continuous with $|F(t,y) - F(t,z)| \le L|y-z|$ for all t, y, z (some $L > 0$).

(a) Show that for $0 < h < \min(1, 1/L)$ the problem $y' = F(t,y)$, $y(0) = y_0$ has exactly one C^1 solution on $[0,h]$.

(b) For $y' = y$, $y(0) = 1$, compute the Picard iterates starting from the constant function 1 and identify their limit.

13. Work in $M_n(\mathbb{R})$ with the operator norm.

(a) Show that $f(A) = A^2$ is differentiable with $Df(A)H = AH + HA$.

(b) Show that $GL_n(\mathbb{R})$ is open and that $g(A) = A^{-1}$ is differentiable on it with $Dg(A)H = -A^{-1}HA^{-1}$.

14. Let $f : \mathbb{R}^2 \to \mathbb{R}^2$, $f(x,y) = (x^2 - y^2,\ 2xy)$. Show that every point $p \neq 0$ has a neighbourhood on which f is a C^1 diffeomorphism onto an open set, but f is not injective on $\mathbb{R}^2 \setminus \{0\}$. What happens at 0?

15. Show that the equation $x^2 + 2y^2 + z^3 + xz = 5$ defines z as a C^1 function $z = g(x,y)$ near $(x,y,z) = (1,1,1)$, and compute $\partial g/\partial x$ and $\partial g/\partial y$ at $(1,1)$.

16. (a) Let $f, g : U \to \mathbb{R}$ be C^1 on open $U \subseteq \mathbb{R}^n$, $M = \{g = 0\}$, and $p \in M$ with $\nabla g(p) \neq 0$. Using the implicit function theorem, show that if $f|_M$ has a local maximum at p, then $\nabla f(p) = \lambda \nabla g(p)$ for some $\lambda \in \mathbb{R}$.

(b) Find the maximum and minimum of xy on the ellipse $\frac{x^2}{4} + y^2 = 1$.

17. Let $f(x,y) = \dfrac{x^2 - y^2}{(x^2+y^2)^2}$ on $(0,1]^2$. Show that

$$\int_0^1 \int_0^1 f(x,y)\,dy\,dx = \frac{\pi}{4}, \qquad \int_0^1 \int_0^1 f(x,y)\,dx\,dy = -\frac{\pi}{4}.$$

Why does this not contradict Fubini's theorem?

18. Prove that $\displaystyle\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$, justifying each use of Fubini and change of variables.

19. Let $D = \{(x,y) : |x| + |y| \leq 1\}$. Using the substitution $u = x + y$, $v = x - y$, compute $\displaystyle\iint_D (x+y)^2 e^{x-y}\,dA$.

Challenge

20. On $U = \mathbb{R}^2 \setminus \{0\}$ let $\omega = \dfrac{-y\,dx + x\,dy}{x^2 + y^2}$.

(a) Show that $d\omega = 0$.
(b) Compute $\int_c \omega$ for $c(t) = (\cos t, \sin t)$, $t \in [0, 2\pi]$, and deduce that ω is not exact on U.
(c) Show that ω is exact on $U' = \mathbb{R}^2 \setminus \{(x,0) : x \leq 0\}$.

21. Let $f : \mathbb{R}^n \to \mathbb{R}^n$ be C^1 with $\|Df(x) - I\| \leq \frac{1}{2}$ for all x. Show that f is a bijection of $\mathbb{R}^n$ and that its inverse is C^1 and Lipschitz.

22. Let $K : [0,1]^2 \to \mathbb{R}$ be continuous and define $(Tf)(x) = \int_0^1 K(x,y)f(y)\,dy$ for $f \in C[0,1]$. Show that T maps $C[0,1]$ into $C[0,1]$ and that the image of the closed unit ball, $\{Tf : \|f\|_\infty \leq 1\}$, has compact closure in $C[0,1]$.

23. (a) Let $R = [a,b] \times [c,d]$ and let P, Q be C^1 on an open set containing R. Using only Fubini's theorem and the fundamental theorem of calculus, prove Green's theorem

$$\int_{\partial R} P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA,$$

where ∂R is traversed anticlockwise.

(b) Explain why (a) and Problem 20(a) do not force $\int_{\partial R} \omega = 0$ for the form ω of Problem 20 when 0 lies in the interior of R.